Chapter 4
Linear Transformations

4.2 Matrix Representations of Linear Transformations

Review: Linear Transformation

Definition

A mapping L from a vector space V into a vector space W is said to be a **linear transformation线性变换** if

$$L(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = \alpha L(v_1) + \beta L(v_2) \quad (1)$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and for all scalars α and β.

Alternative definition

L is a linear transformation if and only if for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and scalars α,

$$L(\mathbf{v}_1 + \mathbf{v}_2) = L(\mathbf{v}_1) + L(\mathbf{v}_2) \quad (2)$$

and

$$L(\alpha \mathbf{v}) = \alpha L(\mathbf{v}) \quad (3)$$

Linear Transformations $\mathbb{R}^n \to \mathbb{R}^m$

If A is any $m \times n$ matrix, and define

$$L_A: \mathbb{R}^n \to \mathbb{R}^m$$

Via

$$L_A(\mathbf{x}) = A\mathbf{x}$$

Then L_A is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Thus, each $m \times n$ matrix defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Question: does each linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ correspond to a $m \times n$ matrix?
What about any linear transformation $L: V \to W$?

Theorem (Standard matrix representation)

If L is a linear transformation mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, then there is an $m \times n$ matrix A, such that for each $\mathrm{x} \in \mathbb{R}^n$,

$$L(\mathbf{x}) = A\mathbf{x}.$$

In fact, the jth column vector of A is given by

$$\boldsymbol{a}_j = L(\boldsymbol{e}_j) \ \ (j = 1,2, \dots, n)$$

Proof

Define $\boldsymbol{a}_1 = L(\boldsymbol{e}_1), \boldsymbol{a}_2 = L(\boldsymbol{e}_2)$,... and let $A = [\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n]$.

If $\mathbf{x} = x_1\boldsymbol{e}_1 + x_2\boldsymbol{e}_2 + \cdots + x_n\boldsymbol{e}_n$ is an arbitrary element of $\mathbb{R}^n$, then

$$L(\mathbf{x}) = x_1 L(\boldsymbol{e}_1) + x_2 L(\boldsymbol{e}_2) + \cdots + x_n L(\boldsymbol{e}_n)$$
$$= x_1 a_1 + x_2 a_2 + \cdots + x_n a_n = A\mathbf{x}$$

Theorem explanation

The theorem tells us how to construct the matrix A corresponding to a linear transformation L. To get the j^{th}column of A, see what L does to the basis element $\boldsymbol{e}_j$ of $\mathbb{R}^n$.

Note in the proof that, just knowing the value of $L(\boldsymbol{e}_1)$, $L(\boldsymbol{e}_2)$,…, $L(\boldsymbol{e}_n)$, by linearity of L we can calculate $L(\boldsymbol{x})$ using matrix-vector multiplication.

Since the standard basis elements $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$ are used for $\mathbb{R}^n$, we refer to A as the **standard matrix representation** of L. **标准矩阵**

Example 1

Find the standard matrix representation for the linear transformation

$L: \mathbb{R}^2 \to \mathbb{R}^4$ given by $L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+y \\ x-y \\ 2x+2y \\ x \end{bmatrix}$.

Solution

- First compute $L(\boldsymbol{e}_1)$ and $L(\boldsymbol{e}_2)$:

$$L(\boldsymbol{e}_1) = L\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\} = (1,1,2,1)^T$$

$$L(\boldsymbol{e}_2) = L\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\} = (1,-1,2,0)^T$$

- Therefore, we have standard matrix representation:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$$

Look, isn't it like the "coefficient matrix" of L?

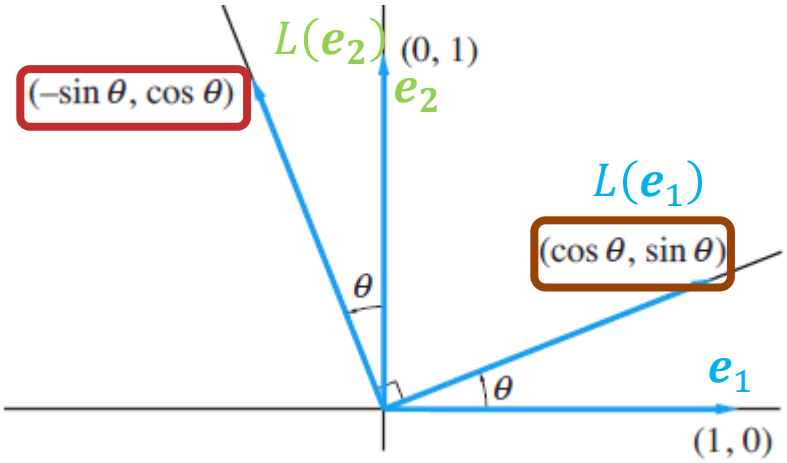
Example 2 (Rotation Matrix)

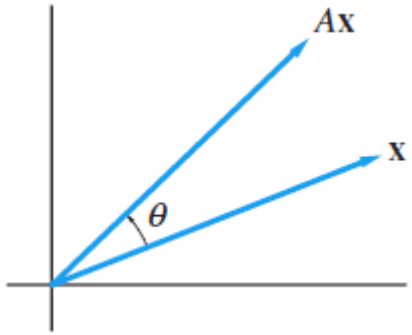
Rotation Matrix for $\mathbb{R}^2$

Let $L: \mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator that rotates each vector by an angle θ in the counterclockwise direction. Then a matrix representation for L is given by

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$L(\boldsymbol{e}_1)$ $L(\boldsymbol{e}_2)$





Matrix Representation Theorem(MRT)

For any linear transformation $L: V \rightarrow W$, after choosing basis for V and for W, all vectors $\mathbf{v}$ in V, and all vectors $\mathbf{w} = L(\mathbf{v})$ in W, can be written as coordinate vectors. Then we can express L as a matrix!

Matrix Representation Theorem

Let $E = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $F = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ be ordered bases for vector spaces V and W, respectively. Then corresponding to each linear transformation $L: V \rightarrow W$, there is an $m \times n$ matrix A such that

$$[L(\mathbf{v})]_F = A[\mathbf{v}]_E \quad \text{for each} \quad \mathrm{v} \in V$$

A is the **matrix representing L relative to the ordered bases** E and F. In fact,

$$\boldsymbol{a}_j = [L(\mathbf{v}_j)]_F \quad j = 1,2, \ldots, n$$

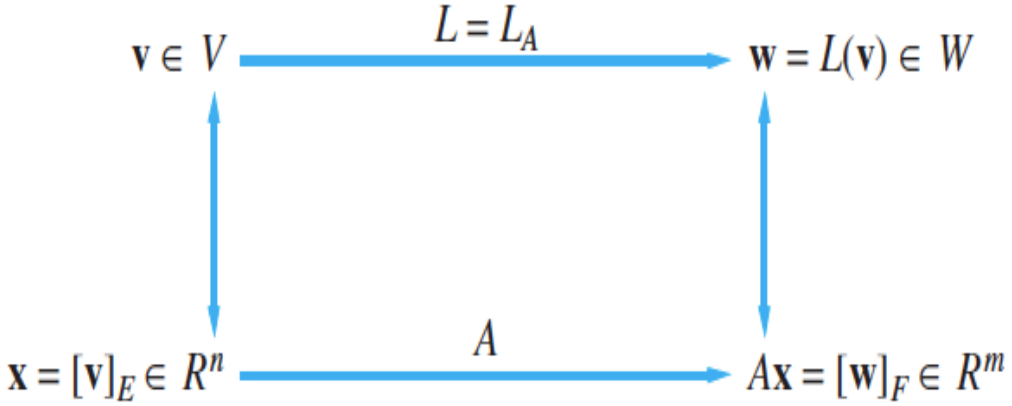
Explanation

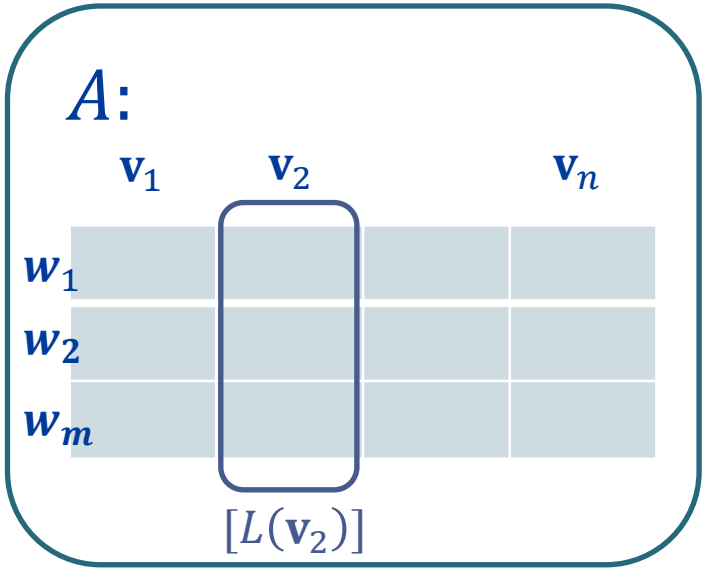
If A is the matrix representing L with respect to the bases E and F, and if

$$\mathbf{x} = [\mathbf{v}]_E \quad \text{(the coordinate of vector } \mathbf{v} \text{ w.r.t. } E \text{)}$$

$$\mathbf{y} = [\mathbf{w}]_F \quad \text{(the coordinate of vector } \mathbf{w} \text{ w.r.t. } F\text{)}$$

Then L maps $\mathbf{v}$ into $\mathbf{w}$ if and only if A maps $\mathbf{x}$ into $\mathbf{y}$.





Key points

- How to write the $m \times n$ matrix A:
  - See what L does to the basis vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in V.
  - Because $L(\mathbf{v}_1)$ is in W, write it as linear combination of $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m$; write the coordinate vector $[L(\mathbf{v}_1)]_F$ as the first column of matrix A.
  - Repeat for $L(\mathbf{v}_2), L(\mathbf{v}_3)$…, to find the 2nd, 3rd,… column of A.
- For any vector $\mathbf{v}$ in V, how to find $L(\mathbf{v})$:
  - Write $\mathbf{v}$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$:

$$\mathbf{v} = \mathrm{x}_1\mathbf{v}_1 + \mathrm{x}_2\mathbf{v}_2 + \cdots \qquad \textbf{So } \mathbf{x} = [\mathbf{v}]_\mathbf{E}$$

  - Because L is a linear transformation:

$$L(\mathbf{v}) = \mathrm{x}_1 L(\mathbf{v}_1) + \mathrm{x}_2 L(\mathbf{v}_2) + \cdots$$

  - The coordinate vector $\mathbf{y} = [L(\mathbf{v})]_F$ is the same as $A\mathbf{x}$

Theorem

Question: How do we calculate the j-th column vector, $\mathbf{a}_j = \left[L(\mathbf{v}_j)\right]_F$?

$$\mathbf{a}_j = \left[L(\mathbf{v}_j)\right]_F \Rightarrow L(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \cdots + a_{mj}\mathbf{w}_m$$

$$= [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m] \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} = W\mathbf{a}_j$$

If $E = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $F = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ be ordered bases for $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $L: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and A is the matrix representing L w.r.t. E and F, then

$$\boldsymbol{a_j = W^{-1}L(v_j)} \quad j = 1,2,\ldots,n$$

Where $W = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m]$.

Example 3

Let L be the linear transformation mapping $\mathbb{R}^3$ to $\mathbb{R}^2$ defined by

$$L(\mathbf{x}) = x_1\mathbf{b}_1 + (x_2 + x_3)\mathbf{b}_2$$

For each $\mathbf{x} \in \mathbb{R}^3$, where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Find the matrix A representing L with respect to the ordered bases $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{\mathbf{b}_1, \mathbf{b}_2\}$.

Solution:

- $L(\boldsymbol{e}_1) = 1\boldsymbol{b}_1 + 0\boldsymbol{b}_2, \;\; L(\boldsymbol{e}_2) = 0\boldsymbol{b}_1 + 1\boldsymbol{b}_2, \; L(\boldsymbol{e}_3) = 0\boldsymbol{b}_1 + 1\boldsymbol{b}_2$
- The jth column of A is determined by the coordinates *of* $L(\boldsymbol{e}_j)$ *w.r.t.* $\{\boldsymbol{b}_1, \boldsymbol{b}_2\}$. Thus

$$A = \begin{matrix} \\ \boldsymbol{b}_1 \\ \boldsymbol{b}_2 \end{matrix} \begin{matrix} L(\boldsymbol{e}_1) & L(\boldsymbol{e}_2) & L(\boldsymbol{e}_3) \end{matrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Corollary

If A is the matrix representing the linear transformation L: $\mathbb{R}^n \rightarrow \mathbb{R}^m$ with respect to the bases $E = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $F = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m\}$ then the reduced row echelon form of $[\,\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m \,|\, L(\mathbf{u}_1), \ldots, L(\mathbf{u}_n)]$ is $[\,I \mid A\,]$.

Remember: this is just trying to write $L(\mathbf{u}_1), \ldots,$ as linear combinations of $\mathbf{b}_1, \mathbf{b}_2 \ldots,$ by solving equations.

Proof

- Let $B = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m]$. The matrix $[B \,|\, L(\mathbf{u}_1), \ldots, L(\mathbf{u}_n)]$ is row equivalent to

$$B^{-1}[B \,|\, L(\mathbf{u}_1), \ldots, L(\mathbf{u}_n)] = [\,I \mid B^{-1}L(\mathbf{u}_1), \ldots, B^{-1}L(\mathbf{u}_n)]$$
$$= [I \,|\, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_n] = [\,I \mid A\,]$$

Example 4

Let $L\colon \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation defined by

$$L(\mathbf{x}) = [x_2, x_1 + x_2, x_1 - x_2]^T$$

Find the matrix representations of L with respect to the ordered bases $\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$, where

$$\mathbf{u}_1 = [1,2]^T, \mathbf{u}_2 = [3,1]^T$$

and

$$\mathbf{b}_1 = [1,0,0]^T, \mathbf{b}_2 = [1,1,0]^T, \mathbf{b}_3 = [1,1,1]^T$$

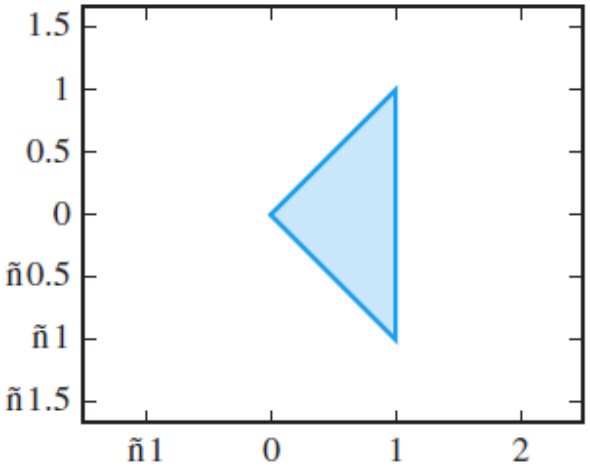
Solution

- First compute $L(\mathbf{u}_1) = [2,3,-1]^T, L(\mathbf{u}_2) = [1,4,2]^T$
- Transform matrix $[\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3 \,|\, L(\mathbf{u}_1), L(\mathbf{u}_2)]$ to the row echelon form:

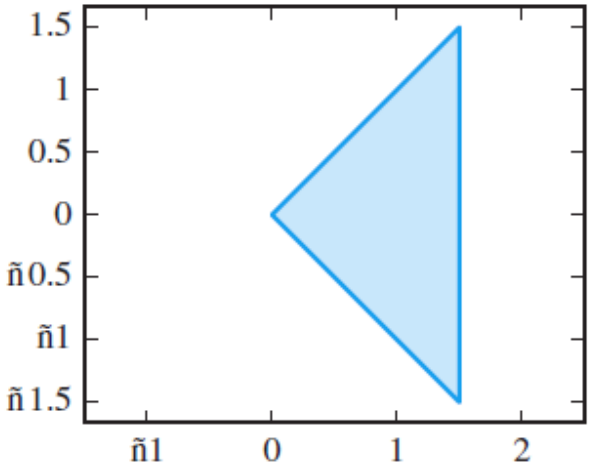
$$\left[\begin{array}{ccc|cc} 1 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 & 4 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|cc} 1 & 0 & 0 & -1 & -3 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right]$$

Matrix representation of L

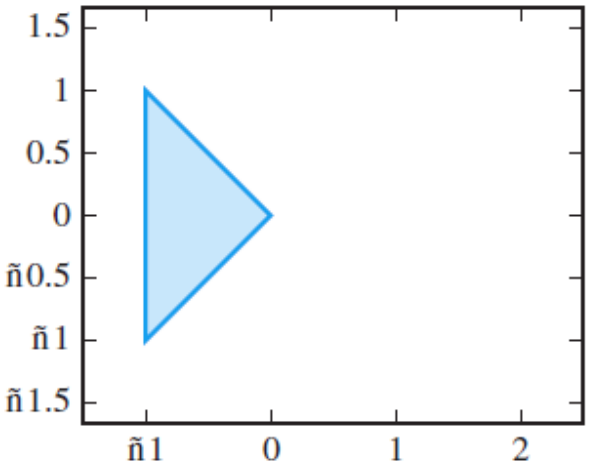
Summary of some common linear transformations in R^2 (Useful in computer graphics and animation)



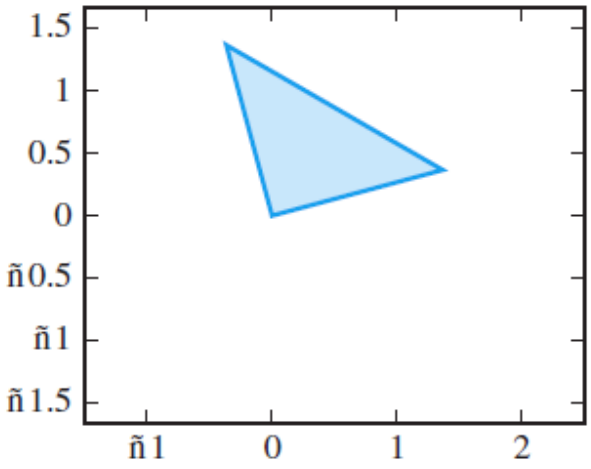
(a) Triangle defined by T



(b) Dilation by factor of 1.5



(c) Reflection about y axis



(d) Rotation by 60°

- Four Basic Transformation Functions
  1. Rotation
  2. Dilation and contraction (Scaling)
  3. Reflection
     - about the y-axis
     - about the x-axis
  4. Translation

What are the transformation functions?

Are ALL of them linear transformations?

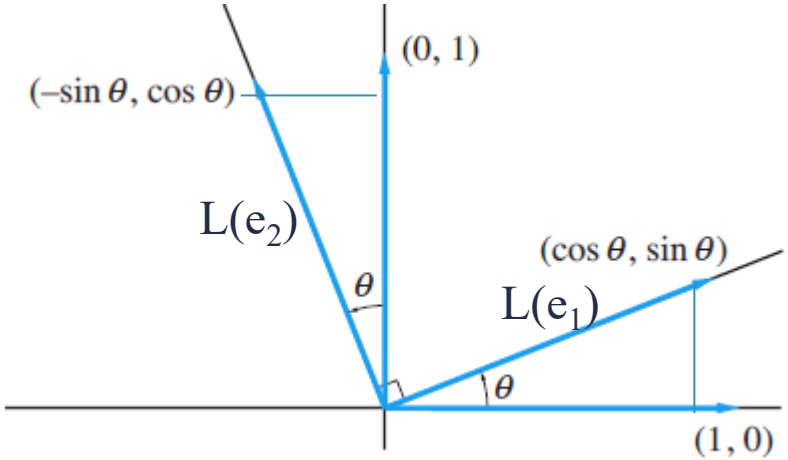
1. Rotation (Already mentioned)

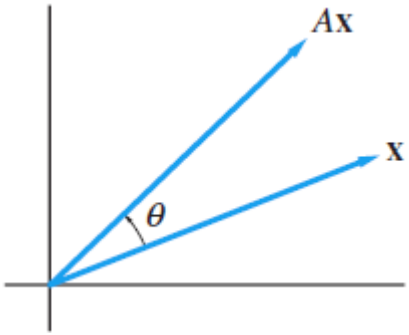
Rotation Matrix for $\mathbb{R}^2$

Let L_R: $\mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator that rotates each vector by an angle θ in the counterclockwise direction.

Then a matrix representation for L_R is given by

$$A_R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$





2. Dilation and contraction (Scaling)

Scaling Matrix for $\mathbb{R}^2$

Let L_S: $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator that dilate or contract any vector by c (scalar in $\mathbb{R}$).

1. Check linearity

$$L_S(x+y) = c(x+y) = cx + cy = L_S(x) + L_S(y)$$

$$L_S(kx) = kc(x) = kL_S(x)$$

i.e. LS is a linear transformation

2. We evaluate $L_S(e_1)$ and $L_S(e_2)$

$$L_S(e_1) = L_S\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}c\\0\end{bmatrix}$$

$$L_S(e_2) = L_S\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}0\\c\end{bmatrix}$$

3. Group the result vectors to form A.

$$\mathrm{A}_S = \begin{bmatrix}c & 0\\0 & c\end{bmatrix}$$

3. Reflection (Mirroring)

3A) Reflection Matrix for $\mathbb{R}^2$ (y-axis)

Let L_{Ry}: $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator that reflect any vector about the y-axis:

To calculate A:

1) We evaluate $L_{Ry}(e_1)$ and $L_{Ry}(e_2)$

$$L_{Ry}(e_1) = L_{Ry}\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-1\\0\end{bmatrix} \qquad L_{Ry}(e_2) = L_{Ry}\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}0\\1\end{bmatrix}$$

2) Group the result vectors to form A.

$$A_{Ry} = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}$$

4. Translation (Transposition)

4) Translation Matrix for $\mathbb{R}^2$

Let $L_T: \mathbb{R}^2 \to \mathbb{R}^2$ be the translation by a vector $\boldsymbol{a} = [a_1, a_2]^2$:

$$L_T(\boldsymbol{x}) = \boldsymbol{x} + \boldsymbol{a}$$

Check linearity:

- $L_T(x + y) = (x + y) + a = x + y + a$, BUT
- $L_T(x) + L_T(y) = (x + a) + (y + a) = x + y + 2a$
- i.e. L is not a Linear Transformation.

WARNING: if a mapping is NOT a linear transformation, there is NO need to calculate a matrix.

(If you are curious about how in computer graphics, one can use matrix to calculate translations, read about "homogenous coordinates" on page 187)